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Unit – 3 : Session – 10 : SLO – 2 | Implementation of Queue Using Linked List — Code

■ I. Node Structure

- struct node { int data; struct node *next; };
- struct queue { struct node *front, *rear; };

■ II. Enqueue Operation

- void enqueue(struct queue *q, int val) {
- struct node *ptr = (struct node *)malloc(sizeof(struct node));
- ptr->data = val;
- ptr->next = NULL;
- if (q->front == NULL) { q->front = q->rear = ptr; return; }
- q->rear->next = ptr;
- q->rear = ptr;
- }

■ III. Dequeue Operation

- int dequeue(struct queue *q) {
- if (q->front == NULL) { printf("UNDERFLOW\n"); return -1; }
- struct node *ptr = q->front;
- int val = ptr->data;
- q->front = q->front->next;
- if (q->front == NULL) q->rear = NULL;
- free(ptr);
- return val;
- }